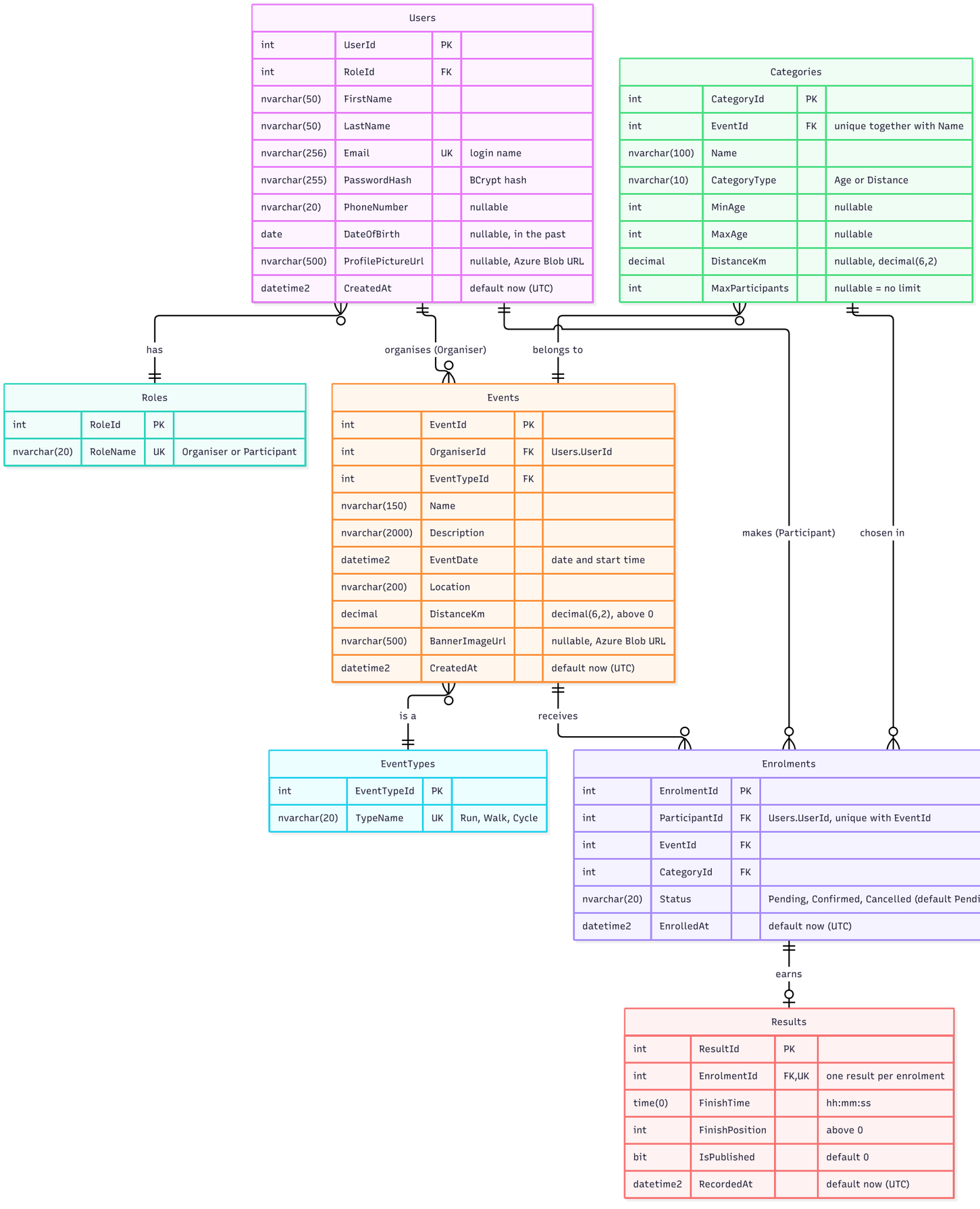
RaceDay – Entity Relationship Diagram

PROG6212 Programming 2B · POE Part 1 · Ryan Naidoo (ST10487315)

Physical ERD – implemented exactly in **docs/RaceDay_Database.sql**



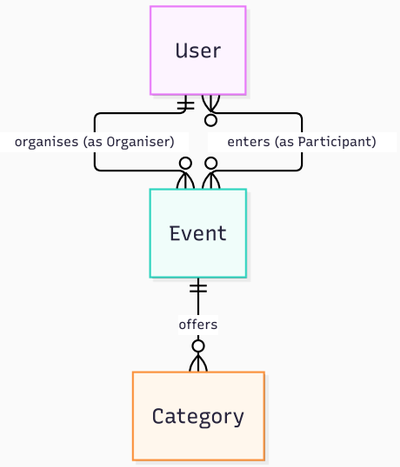
How to read it (crow's-foot notation)

- **exactly one** – the parent end of every relationship
- **zero or one** – an enrolment has at most one result
- **zero or many** – the child end (the table holding the FK)

PK primary key FK foreign key UK unique key

Attributes are **NOT NULL** unless marked *nullable* · types are SQL Server types

Conceptual view: the many-to-many



A participant enters **many** events and an event has **many** participants, which a relational database cannot store directly. The physical ERD resolves this with the **Enrolments** associative entity: one row per entry, which also records the chosen **Category** and links to at most one **Result**.

Relationships in plain English

Each user has one role; a role has many users	`1 : 0..*`
An organiser (user) organises many events	`1 : 0..*`
An event type classifies many events	`1 : 0..*`
An event offers many categories (deleted with it)	`1 : 0..*`
A participant (user) makes many enrolments	`1 : 0..*`
An event receives many enrolments	`1 : 0..*`
A category is chosen in many enrolments	`1 : 0..*`
An enrolment earns at most one result	`1 : 0..1`
Participants and events, resolved by Enrolments	`M : N`

Source: docs/RaceDay_ERD.mmd (Mermaid) · every table, column, key and cardinality is checked against the SQL script by CI on every push

Column details: docs/Data_Dictionary.md